CopC from *Pseudomonas fluorescens* as a model system to explore the copper histidine brace

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Supporting Information

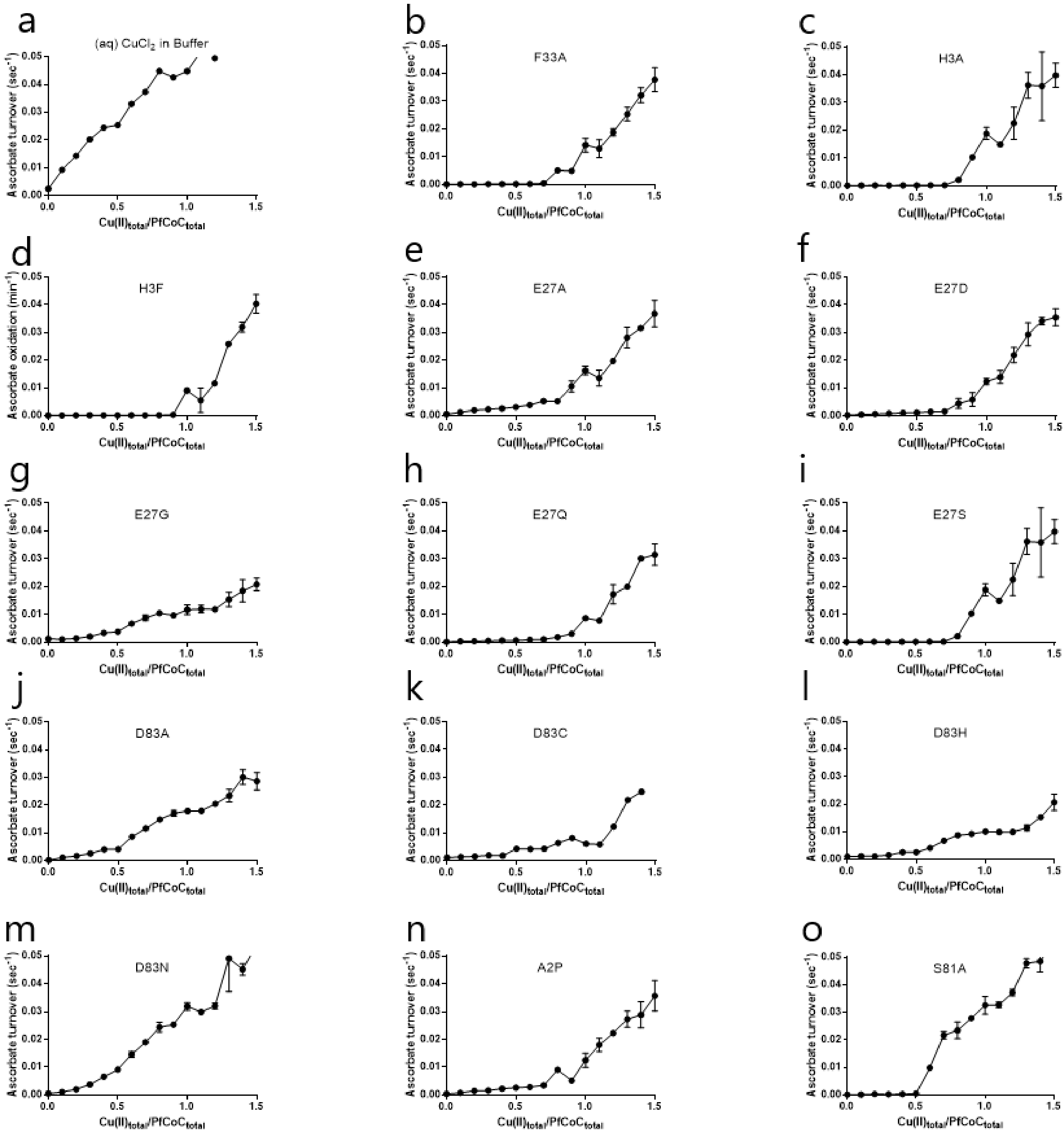

Figure S1 Ascorbate turnover rates for PfCopC variants. CopC variants were incubated overnight with $CuCl_2$ molar equivalent in the range from 0-1.5. Initial ascorbate turnover rates were calculated for each point in the titration and reported in s^{-1}. a) (aq) $CuCl_2$ shows a linear oxidation rate dependency. (b) F33A, (c) H3A and d) H3F do not show any oxidation of ascorbate unless overloading the sample. (e) E27A, (f) E27D, (g) E27G, (h) E27Q and (i) E27S show a small oxidation rate that is dependent on copper concentration but is unlike free copper until the protein is overloaded. (j) D83A, (k) D83C, (l) D83H and (m) D83N show a larger increase in oxidation rates comparable to (aq) Copper (n) A2P shows a no oxidation of ascorbate unless overloading. (o) S81A shows tight binding until a stoichiometry of 1:0.5 and then shows oxidation similar to (aq) copper.

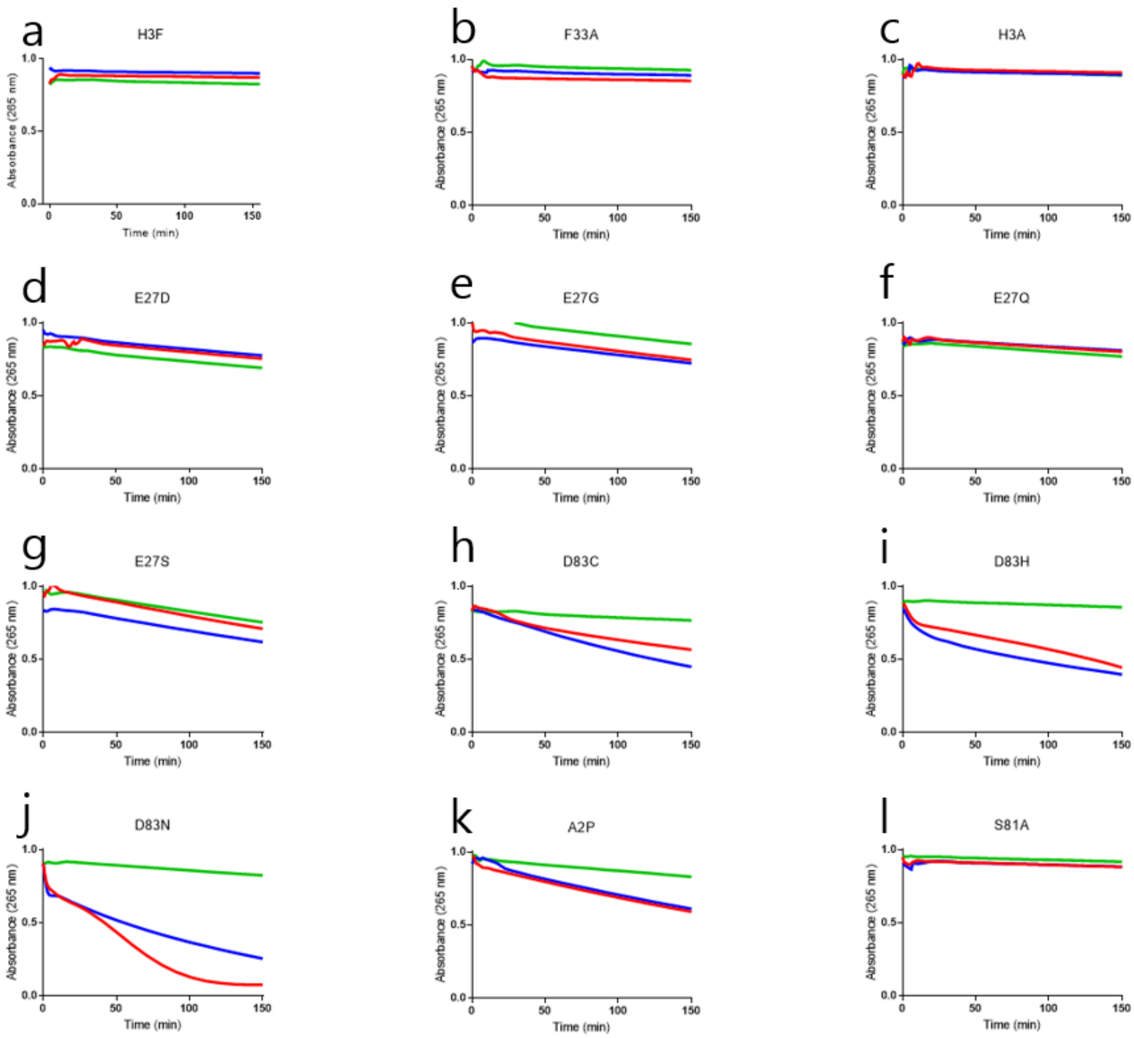

Figure S2. Ascorbate turnover progress curves for PfCopC variants. Shown are data for $CuCl_2$ (red), 1 nM catalase (blue) and 10 µM BCA (green). Signal from 1 µM PfCopC variant equilibrated with 0.5 µM $CuCl_2$ monitored at 265nm for 150 minutes. (a) H3F, (b) F33A and (c) H3A exhibit wild type copper sequestration that is unaffected by scavengers and protects ascorbate from oxidation. (d) E27D, (e) E27G, (f) E27Q and (g) E27S share a slow oxidation profile and are unaffected by BCA and catalase. (h) D83C, (i) D83H and (j) D83N oxidize ascorbate in a time dependent manner, but the oxidations are affected by catalase. BCA quenches ascorbate oxidation. (k) A2P oxidizes ascorbate and is unaffected by catalase but affected by BCA. (I) S81A, behaves identically to PfCopC under these conditions.

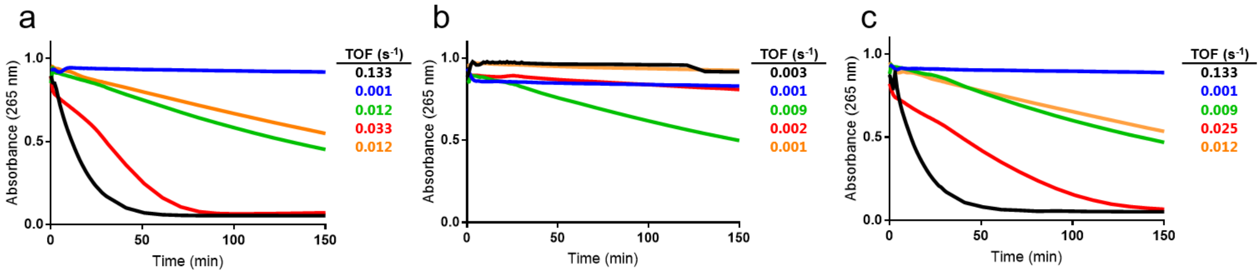

Figure S3 Ascorbate oxidation progress curves for PfCopC and mutants without (a) and with the scavengers (b) 20 µM BCA, and (c) 4 nm catalase. PfCopC (blue), E27A (green), D83A (red), D83H-H85D (orange) were equilibrated with 0.75 µM $CuCl_2$ overnight and together with 0.75 µM $CuCl_2$ (black) tested with 100 µM ascorbate in 20 mM MES buffer pH 6.6.

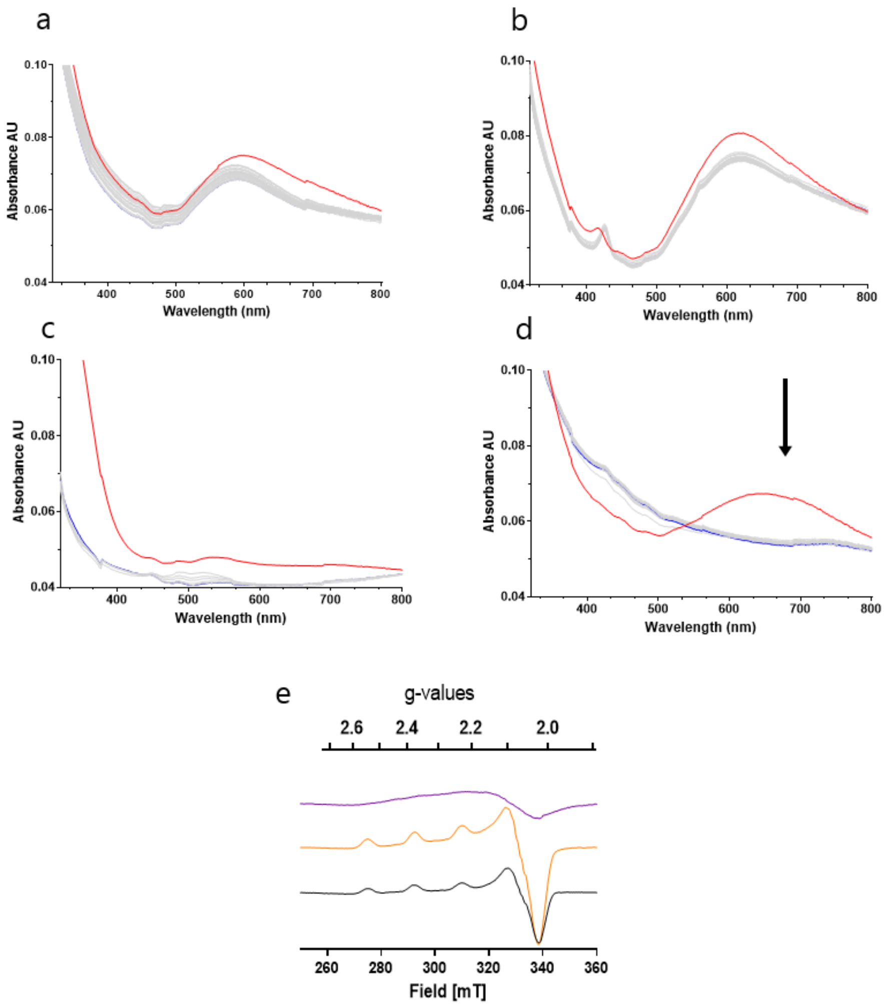

Figure S4. UV-Vis traces of ascorbate reaction with Cu(II)-PfCopC complexes under anoxic conditions. Solutions are 450 µM protein 2mM ascorbic acid, buffered at 6.6 with 20mM MES, 150mM NaCl, the buffer was pH adjusted with HCl. Experiment was monitored at T=0 prior to addition (red line) and in 5 minute intervals for 120 minutes (blue line). (a) PfCopC (b) E27A (c) D83A (d) H85D D83H. (e) EPR spectra of 500 µM D83A (purple) E27A (orange), and E27A after 2 h incubation with 10 mM ascorbate (black). Changes in spectra upon addition of ascorbic acid indicated by black arrow.

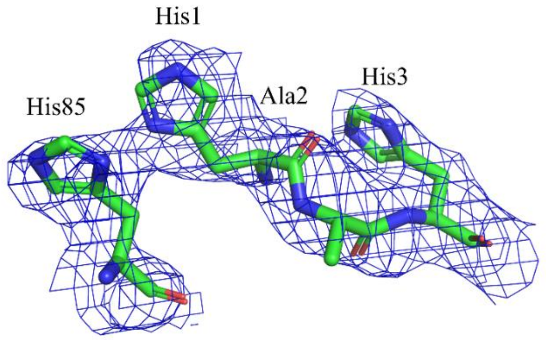

Figure S5 Electron density at the disorded histidine brace of PfCopc D83A chain B. A composite omit map ($2F_o$-F_c) is shown countoured at the 1.0 σ level.

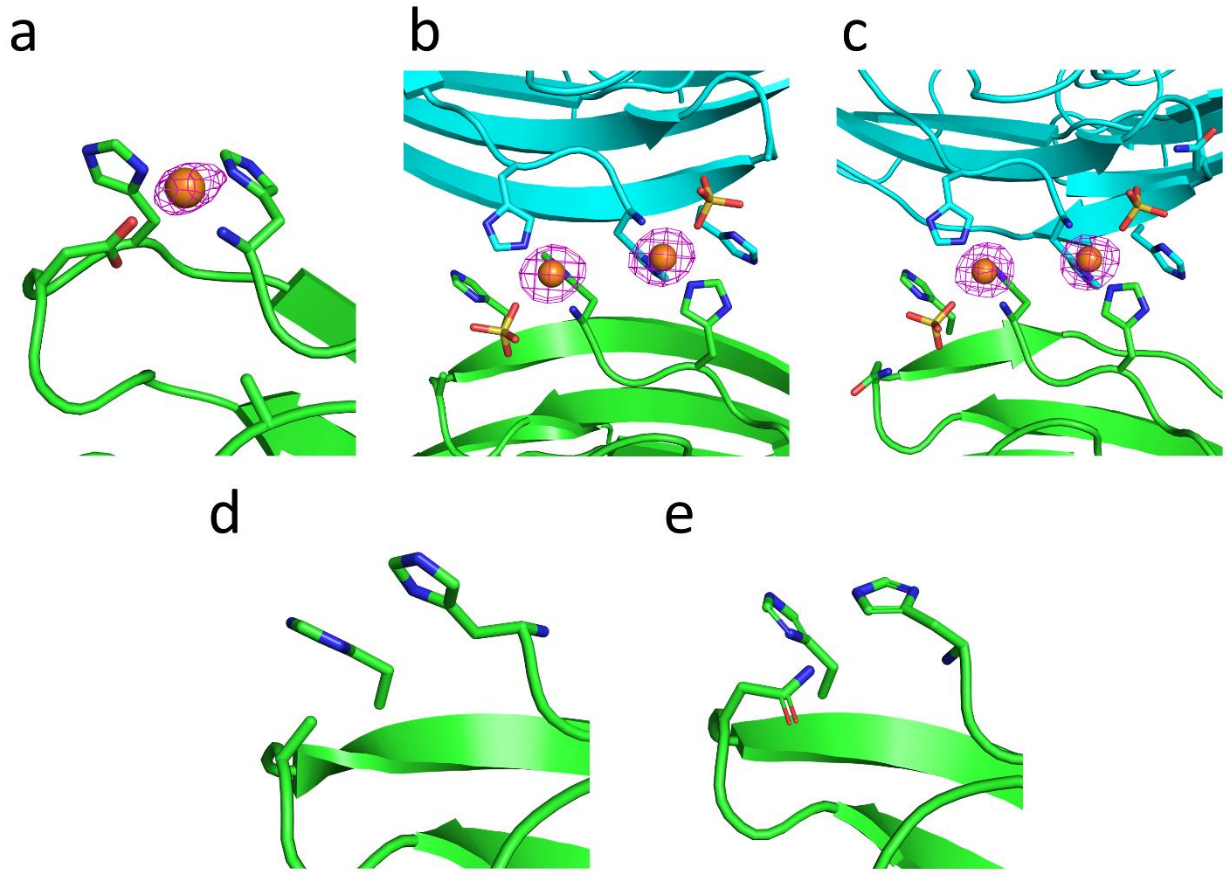

Figure S6 Anomalous difference Fourier maps are shown in Magenta at the His-brace of E27A (a), at the artificial copper binding sites of chain A in D83A (b) and D83N (c), and at the His-brace in chain B of D83A (d) and D83N (e) clearly showing binding at the artificial dimer site, and no binding to the monomeric protein, consistent with the solution studies. The maps are contoured at the 5.0 σ level and carved at 5 Å from the copper atoms in a, b and c and carved at 8 Å from the residues presented as sticks in d and e.

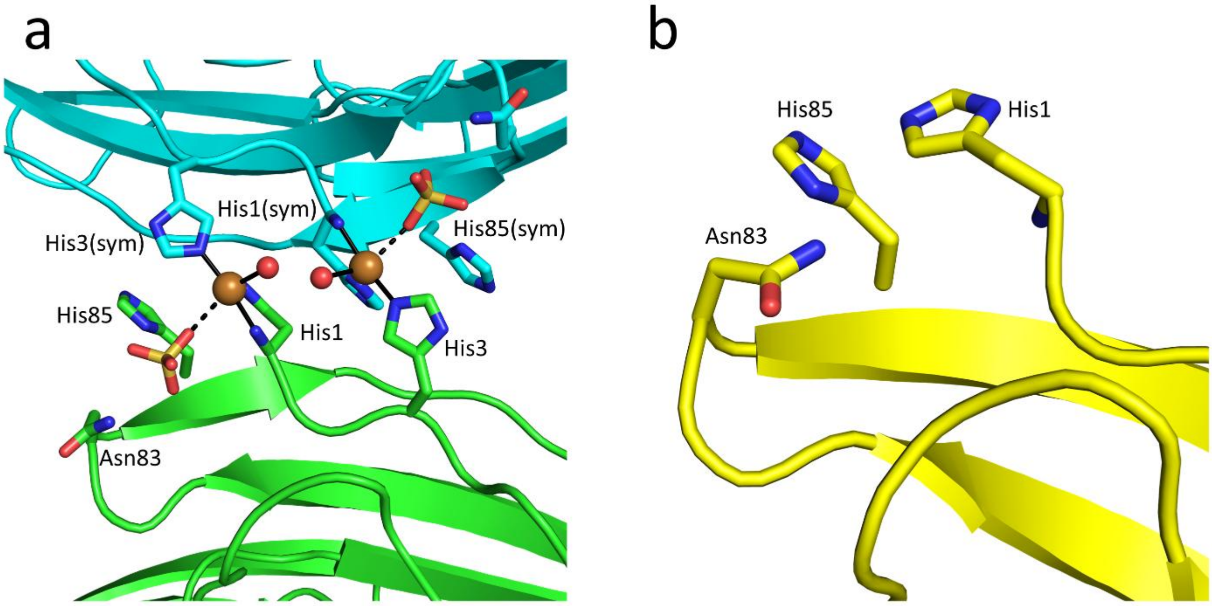

Figure S7 (a) The artificial copper binding site created by the symmetry-related chains A in the structure of D83A and a sulphate from crystallization conditions. (a) His-brace region of D83N chain B, showing that copper is not present at the site.. The site is disordered and exhibits higher B-factors than average.

Table S1 (a) ITC parameters (K_D, ΔH, ΔG and goodness of fit (χ^2) of titrations performed. (a) Ambient. (b) Anoxic. *Apo* proteins where titrated with $CuCl_2$ at ambient conditions and with $[CuI(MeCN_4)]^+$ inside an anoxic environment. Data for PfCopC K_d is taken from [22] in a) and [24] in b).

a

	K_d (M)	ΔH (kcal/mol)	ΔG (kcal/mol)	χ^2
PfCopC	>1.1x10^{-15}	N/D	N/D	N/D
E27A	7.3x10^{-9}	-10.5	-11.1	0.05
D83A	N/A	N/A	N/A	N/A
H85D D83H	N/A	N/A	N/A	N/A

b

	K_d (M)	ΔH (kcal/mol)	ΔG (kcal/mol)	χ^2
PfCopC	1.1x10^{-5}	-11.27	-6.99	0.05
E27A	1.2x10^{-6}	-16.3	-8.09	0.05
D83A	4.9x10^{-7}	27.85	-8.75	2.1
H85D D83H	1.6x10^{-6}	-9.21	-9.32	0.07